Project Charter

for

UNT Parking Availability

Version 1.0

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UNT- CSCE4901.502- Capstone1- Group 3

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1. Project Description

The purpose of this project is to design and implement a Parking Availability App for the University of North Texas that provides real-time information on available parking spaces across campus. The application aims to improve the parking experience for students, faculty, staff, and visitors by reducing the time spent searching for parking, decreasing congestion, and improving overall traffic flow on campus.

UNT Transportation Services has proposed a Parking Availability App that will incorporate ParkZen-style functionality and be embedded into the existing UNT Go app. The application will collect, analyze and display parking data from campus parking lots and garages through a user friendly, map based interface. Users will be able to view current parking availability by lot or garage when arriving on campus.

Key features of the application include:

- Predictive parking analytics based on historical usage patterns
- User assisted data collection to enhance data reliability
- Integration with UNT Go's navigation and campus services
- Interactive, map based visualization of parking availability

2. Business Objectives and Success Criteria

The primary business objective of this project is to enhance the efficiency and effectiveness of campus parking operations by providing timely, accurate, and accessible parking availability information. By reducing the time and effort required for students, faculty, staff, and visitors to locate available parking, the project aims to decrease traffic congestion and improve the overall campus transportation experience. Additionally, the project seeks to increase customer satisfaction with UNT Transportation Services while enabling data driven decision making through the collection and analysis of parking utilization data. Leveraging the existing UNT Go infrastructure will ensure a cost effective implementation and support long term operational sustainability.

Project success will be measured by the system's ability to consistently deliver accurate real-time parking availability information by lot and garage, as well as by measurable be indicated by strong user adoption and satisfaction metrics following deployment, seamless integration with the UNT Go application with minimal service disruption and the availability of actionable parking utilization reports that support operational planning and resource allocation.

3. Stakeholders

Stakeholder	*Major Benefits*	*Attitudes*	*Win Conditions*	*Constraints*
Project Sponsor (Jim Coffey)	Improved usability of premium parking features and foundation for future compass-wide parking availability	Supportive and engaged	Functional and reliable parking availability features integrated into UNT Go	Limited technical background; relies on team and UNT IT for implementation details
Project Team/ Project Manager (Nhan Vu, Thuan Nguyen, Alperen Dalkir, Luis Lozano, Aaditya Yadav)	Successful completion of capstone requirements and hands-on mobile app development experience	Motivated and cautiously optimistic	Project follows the scope, timeline, and academic requirements	Limited development time, academic workload, and varying technical experience
UNTGO user (Students, Faculty, Staff)	Easier access to parking availability information through an existing app	Interested but skeptical	Accurate and easy-to-understand parking information	Dependence on data accuracy and app usability
UNT Permit Holder	Reduce time spent searching for parking	Generally positive	Clear parking guidance that improves daily parking experience	Must download and actively use the UNT Go app
UNT Transportation Services/ IT	Enhanced value of the UNT Go and improved parking communication	Supportive	Stable integration with existing systems	Compliance with university systems, policies, and technical standards.
ParkZen (SDK Provider)	Successful integration and demonstration of SDK capabilities	Neutral to supportive	Proper SDK usage and minimal integration issues	SDK availability, documentation quality, and update timelines

4. Vision

To deliver a streamlined, technology-first parking experience for the University of North Texas that measurably reduce parking search time by Fall 2026

5. Project Scope

In Scope:

- Expand the existing UNT GO mobile app
- Integrate ParkZen SDK into UNT Go application
- Display real-time parking availability on a map-based interface
- Tracking of two primary parking activity phases: student arrival and student departure
- Retrieval of parking data from two data sources using APIs and SDKs
- Processing and visualization of parking availability data
- Develop a user interface components to display available parking by lot or garage

Out of Scope:

- Building a new mobile application from scratch
- Major front-end redesign or custom visual branding changes
- Installing or managing physical hardware (cameras, sensors, etc.)
- Development, modification, or maintenance of the ParkZen SDK
- Enforcement of parking policies or ticketing functionality

6. Major Deliverables

Deliverable	***Description***
Integrated Parking Feature	Real-time display of parking availability within the existing UNT Go mobile application.
Map-Based User Interface	Interactive visualization of parking lots showing available spaces.
ParkZen SDK Integration	Functional connection to ParkZen APIs to retrieve parking availability data.
Data Processing and Analytics	Backend logic to collect, process, and display parking data accurately.

Minimum Viable Product (MVP) Prototype	Working version of the app demonstrating core functionality.
System Testing and Validation	Thorough testing to ensure reliability, usability, and integration with UNT Go.
Final Project Demonstration and Documentation	Complete demonstration of the project to stakeholders, along with technical documentation and training materials.

7. Assumptions and Dependencies

AS-1	ParkZen will provide SDK access and development documentation.
AS-2	The mobile framework solution will support Android and iOS.
AS-3	Jim Coffey will connect us with additional stakeholders as needed.
AS-4	ParkZen SDK provides reliable data for parking availability.
AS-5	The UNT GO codebase will be accessible and modifiable by our team.
DE-1	UNT GO app codebase and documentation must be shared with the team before development can begin.
DE-2	ParkZen SDK must be provided before integration can begin.
DE-3	API and SDK access to both databases is required for pulling parking data.
DE-4	Mobile framework selection must be finalized before development starts.
DE-5	Collaboration with the UNT GO team for SDK support and troubleshooting.

8. Constraints

Dimension	*Constraint (state limits)*	*Driver (state objective)*	*Degree of Freedom (state allowable range)*
Features	Must rely on existing parking systems (AIMS, LPR/Genetec, Premium Parking, ParkZen). No physical sensors added. Parking space reservation is not supported.	Deliver parking availability that provides real-time visibility of available spaces through "white label" in UNT GO.	Feature set may vary based on data availability; Release 1.0 focuses on availability display only, with crowdsourcing, prediction, and advanced analytics deferred to later releases.
Quality	Data is not fully real-time due to scan intervals, transaction timing, and early-stage lack of crowdsourced data. Availability cannot be guaranteed.	Deliver useful, trustworthy availability estimates that improve with increased data sources and user participation.	Accuracy may range from near real-time to short-delay estimates. The system would improve quality over time through the use of crowd-sourcing adoption.
Cost	No additional physical sensors or fixed cameras may be installed. Most resources are ready, but could be subject to change.	Minimize implementation and maintenance costs by leveraging existing infrastructure.	Budget flexibility applies only to software integration and configuration. Hardware expenditures are fixed at zero beyond existing LPR systems.
Schedule	Development depends on integration readiness and data access of third-party vendors.	Deliver a functioning product within intended academic deadlines.	Scope and data sources may be phased. The initial release may rely primarily on LPR and premium parking data, with crowdsourced inputs added post.
Staff	A team of 5-6 developers/pro duct managers	Successfully integrate the multiple systems and deliver a working solution	Responsibilities may be redistributed within the team. Complexity of algorithms and prediction models may be reduced to fit team capacity.

9. Milestones

Event or Deliverable	*Target Date*	*Responsibility*
Project charter approved	Feb 01, 2026	Project Sponsor
Project plan completed	Feb 15, 2026	Project Manager
Project plan approved	Feb 22, 2026	Project Sponsor
Project team assembled	Feb 22, 2026	Project Team/ Development Team
Requirements validated	Feb 15, 2026	Project Manager
Project execution initiated	Feb 23, 2026	Project Manager/ Development team
MVP Prototype completed	TBD	Development Team
System testing completed	TBD	Development Team
Project execution completed	TBD	Project Team/Development Team
Customer acceptance	TBD	Project Sponsor/Users
Final Project Demonstration	TBD	Project Team
Project closed out	TBD	Project Manager

** Due to the structure of the Capstone project, the same individuals serve as both the Project Manager and the Development Team.*

10. Business Risks

Risk	*Probability*	*Impact*	*Mitigation*
System outright fails	Low	Critical	Conduct a detailed system testing, integration testing, and stakeholder review before deployment
System encounters bugs	Medium	Medium	User iterative development, regular testing cycles, and issue tracking to identify and resolve defects effectively and early.
Inaccurate parking data	Medium	High	Verify data from the SDK and refresh data frequently to ensure accuracy
Delays in ParkZen and SDK availability or updates	Medium	Medium	Use sample or mock data until the SDK is available.
Dependency on ParkZen SDK	Medium	High	Work closely with ParkZen and follow their integration guidelines, design a fallback behavior when data is unavailable.
Low user adoption or engagement	Medium	High	Involve stakeholders early, gather feedback during development and ensure the interface is simple and intuitive.
Requirement changes (parking policies)	Medium	Medium	Maintain regular check-in with project sponsors and manage scope changes as needed.
Academic schedule constraints (exams, other courses or semester deadlines)	High	Medium	Create a realistic timeline, prioritize high-value features, and maintain clear task ownership.

11. Resources

Resource	*Description and Source*
Project Team	Capstone Group 3 consists of five student developers responsible for analysis, design, implementation, testing, and documentation.
Project Sponsor	Jim Coffey, providing project vision, stakeholder coordination, and approval guidance.
UNT Go Application	Existing UNT Go mobile application codebase provided by UNT IT - Transportation Service for integration and enhancement.
ParkZen SDK	Third-party SDK provided by ParkZen to retrieve real-time parking availability.
APIs & Databases	APIs and SDK interfaces used to access parking availability and related data from two data sources.
Development Tools	Android Studio, Xcode, version control tools, and testing tools required for mobile app development.
Mobile Framework	Cross-platform or native mobile framework to be finalized and provided prior to development.
Testing Devices	Android and IOS mobile devices or devices or emulators for functional and integration testing.
Documentation and Training	ParkZen SDK documentation, UNT Go technical documentation, and stakeholder guidance as needed.
University Facilities	UNT computing resources, collaboration tools, and meeting spaces to support development and coordination.

12. Approvals

Approval Decision:

□ Approved, development of detailed project plan is authorized
□ Approved, project execution is authorized
□ Approved, but project is on hold until future notice
□ Revise charter and resubmit for approval
□ Charter and project proposal are rejected

Role or Title	*Name and Signature*	*Date*
Project Sponsor	Jim Coffey	

Revision History

Name	*Date*	*Reason For Changes*	*Version*